

To: Dr. Pengcheng Xi, Dr. Chang.Shu

Week 4 Report

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NRC - Digital Technology, 04.02.2026

1. Work Completed Since Last Meeting

This week, I focused on implementing the “marker fitting” algorithm essential to our data processing pipeline. However, there have been challenges implementing MoSh++ - the state-of-the-art solution that we planned to use. Therefore, a careful study was conducted and a plan was made to pivot to a custom VPoser-based pipeline that maintains the statistical prior of MoSh++ while running on modern PyTorch infrastructure.

1.1. The MoSh++ Saga & Pivot

Initial Plan: Deploy MoSh++ (the industry standard for AMASS) to replace the broken legacy pipeline.

Outcome: After significant effort containerizing the application, I discovered a fundamental code incompatibility between the legacy chumpy solver (2016) and modern SMPL model formats.

Decision: Rather than patching the massive unmaintained codebase, I decided to pivot to a **Custom VPoser Pipeline**. This extracts the “Brains” of MoSh++ (the statistical prior) but runs it on a modern PyTorch engine.

1.2. Why the Legacy Code Failed (The “Inflation” Bug)

The previous custom fitting pipeline uses a heuristic optimizer to fit markers directly to joint centers (bones).

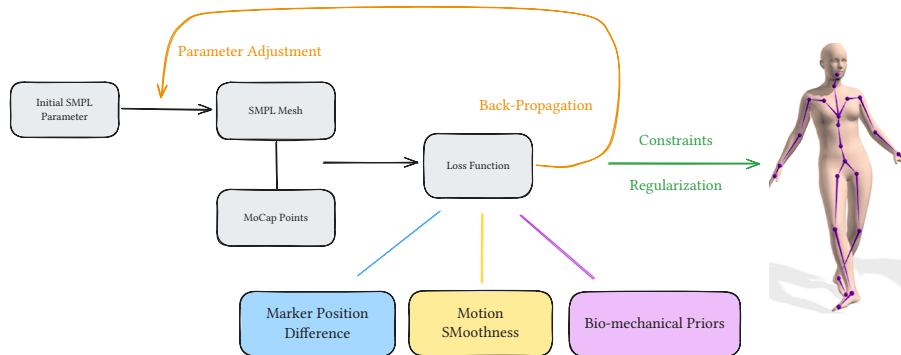


Figure 3: The previous pipeline using custom backpropagation optimization.

It produced meshes that appeared “inflated” or bloated. This is because it **mapped skin-surface markers (e.g., an Elbow marker) directly to the internal Joint Center (Bone).**

Consequence: To satisfy this constraint, the optimizer forced the bone to expand outwards until it touched the skin surface, artificially widening the subject.

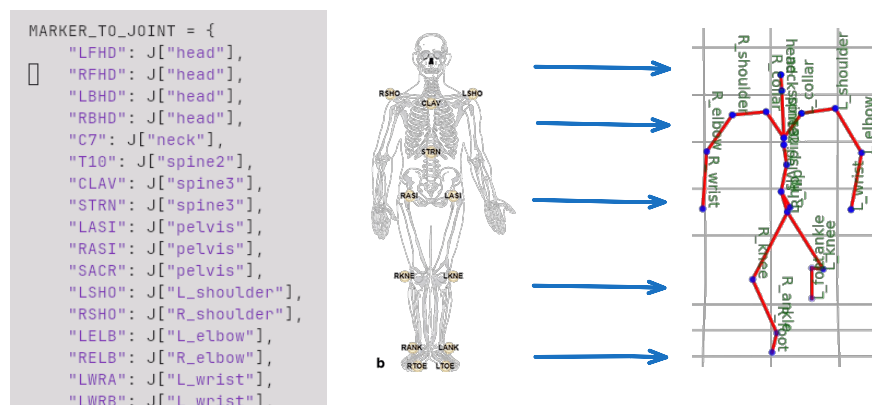


Figure 1: The legacy approach incorrectly mapped skin markers to internal joint centers. This forced the optimizer to “inflate” the skeleton to minimize error.

1.3. Correct Approach (Vertex Mapping)

Solution: We must map markers to specific **Vertex IDs** on the mesh skin.

Benefit: This allows the bone to remain in the correct anatomical position while rotating to align the skin surface with the marker.

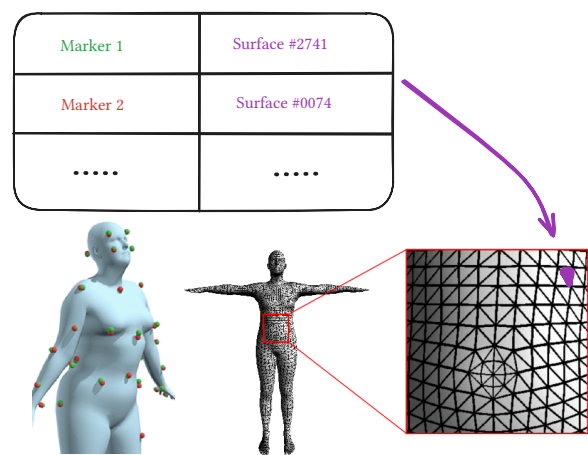


Figure 2: The corrected approach maps markers to specific mesh vertices (e.g., L_Elbow_Marker → Vertex 2741). This preserves correct body volume.

1.4. New Architecture (Custom VPoser Pipeline)

We are replacing the broken MoSh++ container with a custom script that utilizes the VPoser library directly.

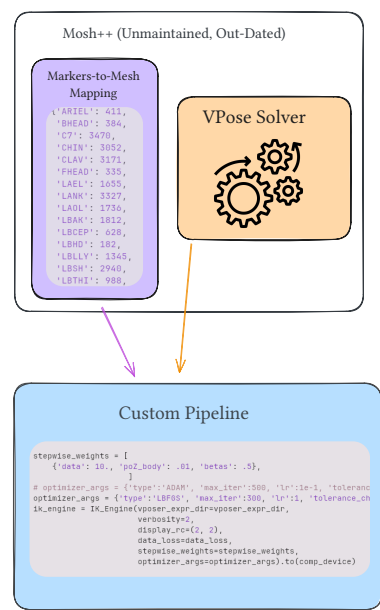


Figure 4: The new architecture replaces the unmaintained MoSh++ legacy stack with a direct PyTorch implementation using the VPoser IK engine.

1.5. Comparison of Pipelines

	Legacy Pipeline (Eugene)	MoSh++ (Standard)	Proposed Custom Pipeline
Marker Mapping	Direct to Joint (Incorrect)	Full Vertex Mapping	Full Vertex Mapping
Solver Engine	Custom Heuristic	VPoser Solver Engine	VPoser Solver Engine
Codebase Status	Produces “Inflated” data	Broken (Dependency Hell)	Maintained PyTorch Ecosystem

2. Extended Abstract Strategy

As we are ten days close to the Feb.15th deadline to submit an extended abstract to the 3DV conference, it’s good to start drafting. I have prepared three alternatives for the upcoming abstract submission, supervisor feedbacks on the preferred framing will be very helpful.

Alternative 1: Intermediate	Alternative 2: Proof-of-Concept	Alternative 3: Methodology
Title: Towards Age Controllable Motion Generation	Title: Can AI Learn Aging? Kinematic Signatures in Diffusion	Title: Validating Subtle Motion Style with ST-GCN
Short Abstract: This research develops an age-conditioned motion diffusion model using Low-Rank Adaptation (LoRA). We fine-tune a base diffusion model on motion capture data segmented by age (Young, Mid, Elderly). By interpolating between age-specific LoRA adapters, we achieve continuous age control in generated walking motions. My Assessment: High Risk. Continuous aging is subtle and difficult to prove visually in a short time-frame.	Short Abstract: This study investigates whether AI can learn age-related kinematic signatures from human motion. We train a Spatial-Temporal Graph Convolutional Network (ST-GCN) to classify age from motion capture data and develop a diffusion model with age-conditioned generation. My Assessment: High Robustness. Success defined by ST-GCN classification accuracy (quantitative), reducing reliance on perfect visual generation.	Short Abstract: We propose using Spatial-Temporal Graph Convolutional Networks (ST-GCN) as objective validation metrics for Low-Rank Adaptation (LoRA) of motion diffusion models. Unlike human observation which detects obvious style changes, ST-GCN enables validation of subtle adaptations like age. My Assessment: High Risk. The metric itself must perform impressively to justify the paper.

Recommendation: Alternative 2 (Proof-of-Concept) is recommended. This frames the project as exploratory research validating whether AI can learn age-related kinematic signatures. It investigates **if** AI can decouple age from motion using ST-GCN to validate the “Age Signal” exists, then uses Diffusion to generate it. Success is defined by classification accuracy rather than perfect visual generation.

3. Project Timeline & Adjustments

Week 1	Onboarding, foundational study, project structure (Completed)
Week 2	Foundational study; dataset analysis; pipeline investigation (Completed)
Week 3	Pipeline audit; MoSh++ containerization (Completed)
Week 4	Age-based data splitting (Completed), Pipeline diagnosis (Completed), VPoser pivot implementation (New)
Week 5	Complete VPoser implementation; generate Golden Dataset; Train example ST-GCN Network
Feb 15	Submit extended abstract (Target)
Week 6-7	Train LoRA Adapters; Fine-tuning; validation experiments; metrics
Week 8-9	Full paper writing; supplemental experiments

Red indicates postponed tasks; **Purple** indicates revised tasks.

Timeline remains tight but achievable. The VPoser pivot represents a more robust long-term solution despite the additional development time required.